

Qatar University
 College of Arts and Sciences
 Department of Mathematics and Statistics

STAT 481 — R Laboratory 1

Discover a Data Matrix of Your Own — from a public source to a technical report

Out: **Thu, Sep 3, 2026** • Due: **Thu, Sep 17, 2026** by 23:59 (Blackboard)

Topics: *everything in Lectures 1–5: the data array, $\bar{x}$, **S**, **R**, $|\mathbf{R}|$, displays, distances*

Instructor: Prof. Dhafer Malouche • Individual work • Total: 20 pts

What this lab is. Every data set you have met so far was handed to you: `USairpollution` in Lecture 1, `worldhealth2022` and the 52-column `whwide2022` panel from Lecture 2 onward. This time nobody hands you anything. You go to one of three public sources, decide what the *rows* are and what the *columns* are, download the numbers, build the $n \times p$ data matrix yourself, and then do to it everything Lectures 1–5 taught you to do to a data matrix — ending in a **technical report** written in Quarto that a reader could re-run from the first line to the last.

The report is the deliverable. The data matrix you build is the first thing it is graded on, and it is the part nobody can do for you. Budget twelve to fifteen hours over the two weeks: four or five for finding and building the data (the part that cannot be rushed), five for the analysis, three for the writing and the fresh-session render. Start the download in the first week — servers, packages and missing values all take longer than they look.

What you submit — three files, nothing else

1. `RLab1_Lastname.Firstname.ID.qmd` — your Quarto source.
2. `RLab1_Lastname.Firstname.ID.pdf` — the rendered output of *that* source.
3. `RLab1_Lastname.Firstname.ID.data.csv` — the data matrix your download code built, exactly as your report reads it.

The reproducibility rule. I render your `.qmd` myself, with your CSV beside it. A report whose source does not render, or that renders to different numbers or figures from the PDF you submitted, **loses all reproducibility marks** (a different package version in `sessionInfo()` is not a difference). I also run your download chunk on its own: it must rebuild your CSV (the World Bank, CDC and NCI servers can change a value between your download and mine — a difference in the third decimal is not a fault; a chunk that does not run is).

Choose one topic

Each topic names a source, says what one *row* is, and sets the same floor: $n \geq 40$ **rows**, $p \geq 8$ **numeric columns**, in **at least two different units** (so that Lecture 5’s “standardise or not” question has teeth), and **nothing that appears in `worldhealth2022` or `whwide2022`**. A column that carries a *label* (a region, a sex, a census division) is not one of the eight, but it is worth having — Task 4 rewards it. Within the floor, *which* rows and *which* columns are yours to choose, and Task 1 grades the choice.

Topic A — Countries: the World Bank (two doors)

A1. World Development Indicators. One row per country, **one year for every column** (2021 or 2022 — probe both before choosing; coverage differs by indicator, and the year with the fewest missing cells across your eight wins). Choose eight or more indicators from families the course panel does *not* use — energy and environment, education and gender, labour, ICT

and infrastructure, finance and business, governance — and check each one's coverage before you commit to it. The API needs no key and no package:

```
library(jsonlite) # fromJSON(): read a web address that returns JSON
u <- paste0("https://api.worldbank.org/v2/country/all/indicator/",
            "EG.ELC.ACCS.ZS?date=2021&format=json&per_page=400")
d <- fromJSON(u)[[2]] # element 1 is paging info; 2 is the data
str(d)                # countryiso3code, country$value, date, value
```

Indicator codes are on <https://data.worldbank.org/indicator> (the code is the end of each page's address). **Trap to discover:** the list is *not* a list of countries — it also contains the Bank's regional and income aggregates ("World", "Arab World", "High income"), and a data matrix that keeps them is wrong from the first row. https://api.worldbank.org/v2/country?format=json&per_page=400 tells you which is which (its `region$value` reads "Aggregates" for those rows — and it also hands you a region label for the real countries).

A2. Enterprise Surveys. One row per economy: the World Bank's firm-level surveys summarised to country indicators — corruption, crime, infrastructure, finance, workforce, trade — 115 indicators, about 146 economies. They live in the same API under source 13; the indicator list is at https://api.worldbank.org/v2/sources/13/indicators?format=json&per_page=200 and a series is fetched with `?source=13&format=json&per_page=20000` appended to the same address form as A1 (no `date=`). **Trap to discover:** each economy was surveyed in a *different year* (2006 to 2020 in the API), so one indicator comes back as several rows per economy, most of them NA. You must decide how to build one row per economy — keeping the *latest* survey is the usual choice: `order(iso3, -year)`, then each economy's first row, `!duplicated(iso3)` — and say in the report what your choice costs (a 2006 survey and a 2020 survey are now one row each).

Topic B — People: NHANES 2021–2023 (CDC)

One row per adult participant of the US National Health and Nutrition Examination Survey, cycle L. The `nhanesA` package downloads one table per call; the tables share the key `SEQN`:

```
library(nhanesA) # install once, in the console
demo <- nhanes("DEMO_L") # demographics: RIAGENDR (sex), RIDAGEYR (age)
bmx <- nhanes("BMX_L") # body measures: BMXWT, BMXHT, BMXBMI, BMXWAIST
bpx <- nhanes("BPXO_L") # blood pressure: BPXOSY1, BPXODI1
tch <- nhanes("TCHOL_L") # total cholesterol: LBXTC
hdl <- nhanes("HDL_L") # HDL cholesterol: LBDHDD
N <- merge(demo[, c("SEQN", "RIAGENDR", "RIDAGEYR")], bmx, by = "SEQN")
```

Other tables (glucose `GLU_L`, bone density, physical activity, ...) are listed at <https://wwwn.cdc.gov/nchs/nhanes/> under 2021–2023. Restrict to adults (`RIDAGEYR ≥ 20`); keep `RIAGENDR` as your label. **Traps to discover:** each merge *loses* rows (who was measured but not bled?); the missing-value count differs by column and the report must say what you did about it; and the *label* columns (`RIAGENDR`, race, education) arrive as `Factors` with words in them (`str()` tells you) — keep them out of the matrix you hand to `cov()`, and choose your participant by a rule you state (the adult with the median BMI, or one seeded random draw).

Topic C — Places: cancer incidence from SEER / NPCR (NCI)

One row per US state (plus DC and Puerto Rico), one column per cancer site: the age-adjusted incidence rate per 100 000, 2018–2022, as published by the NCI's *State Cancer Profiles* from the SEER and NPCR registries. Every table is a web address that returns a CSV; the site code is the number after `cancer=`:

```
u <- paste0("https://statecancerprofiles.cancer.gov/incidencerates/",
            "index.php?stateFIPS=00&areatype=state&cancer=047&race=00&sex=0",
```

```

"&age=001&stage=999&year=0&type=incd&sortVariableName=rate",
"&sortOrder=default&output=1")      # 047 = lung & bronchus
txt <- readLines(u, warn = FALSE)    # one string per line of the file
head(txt, 12)                        # look before you parse

```

Site codes (from the page's own menu): 001 all sites, 020 colon & rectum, 035 liver, 040 pancreas, 047 lung, 053 melanoma, 055 breast (female: `sex=2`), 057 cervix, 061 ovary, 066 prostate (`sex=1`), 071 bladder, 072 kidney, 076 brain, 080 thyroid, 086 non-Hodgkin lymphoma, 090 leukemia. You may add columns by sex, age band (`age=136` for 50+) or race, and `areatype=county` inside one state gives a county-level version with far more rows. **Traps to discover:** the file is not a clean CSV — a title, blank lines, a header, the rows, then footnotes; state names carry a footnote mark (“Kentucky(7)”); a national total row sits among the states; suppressed cells print as *. Parsing one file is four lines — `grep("^State,FIPS", txt)` finds the header line, the rows run to the next empty line, `read.csv(text = ...)` reads that slice, `sub()` cleans the names — and a `for` loop over the site codes does the rest. **The two-unit floor here:** every rate is per 100 000, so add *one or two* columns in another unit from the same files — the *Average Annual Count* for all sites (cases per year), or a rate for one age band — not one count per site, or *p* climbs towards *n*. Everyone in Topic C will have nearly the same columns; that is expected, and your subset, your order and your unit of interest are where the choices are. A census region typed by hand for 52 rows is a legitimate label.

How to work

Open the course project, create your `.qmd` in it, and keep **two kinds of chunk apart**: the *download* chunk(s), which talk to the internet and end with `write.csv(..., row.names = FALSE)` to your CSV; and the *analysis* chunks, which start from `read.csv("RLab1...data.csv")` and never touch the internet. Three chunk options Lecture 3 did not need, each a first line inside the chunk: `#| eval: false` tells Quarto to *show* the code without running it and `#| echo: true` makes sure it is shown — give the download chunk both once the CSV exists; `#| fig-cap: "..."` puts a caption under the chunk's figure. A table in prose is a Markdown table (a row of `|`-separated cells, a row of `|---`, then the rows); a printed data frame is also a table. Everything else renders live. Set the row names of your data matrix to the unit's name or code, so that every distance in Task 4 prints with a name attached.

Only the R you have been taught, plus the download helpers. Everything in Lectures 1–5 is allowed: `dim`, `str`, `head`, `[]`, `$`, `rownames`, `sort`, `order`, `rank`, `colMeans`, `apply`, `tapply`, `cov`, `cor`, `det`, `hist`, `plot`, `corrplot`, `ggpairs`, `replicate`, `rnorm`, `set.seed`, `scale`, `dist`, `as.dist`, `pheatmap`, `mahalanobis`, `solve`, `sweep`, `rowSums`, `complete.cases`, `is.na`, `t`. Ten you have not typed before, all for building the matrix: `paste0(a, b)` glues strings together with nothing between them (long web addresses are built this way); `fromJSON(u)` reads a web address that returns JSON; `readLines(u)` reads one that returns text, one string per line; `read.csv(text = x)` parses a character vector as if it were a file; `merge(a, b, by = "id")` joins two tables on a shared key and keeps only the rows present in both; `match(a, b)` gives the position in `b` of each element of `a` (so `y[match(x$iso3, y$iso3)]` lines a second table up with the first); `sub(pattern, "", x)` deletes the first match of a pattern from each string — `"\\([0-9]+\\)$"` is “a bracketed number at the end”; `duplicated(x)` is `TRUE` for every element that has already appeared earlier in `x`; `sample(x, 40)` draws 40 of `x` without replacement (seed it); and `write.csv(d, file, row.names = FALSE)` writes the CSV. Base R you already use is always fine — `median`, `sd`, `sum`, `which`, `colSums`, `as.numeric`, for loops, `function(x)` inside `apply` — the rule is about *methods*: **no test, no p-value, no principal component, no clustering, and not the word “outlier”**: none of them has been defined yet. Say “unusual”.

Collaboration & AI tools. You may discuss ideas with classmates, and you may use AI tools *as a study aid* (checking reasoning, exploring R syntax, decoding an error). You must **write your own report in your own words, build your own data matrix** (in Topics A and B, two students with the same columns from the same source in the same year will both be asked to explain why), and **acknowledge**

any such use in a line at the end. You may be asked to explain any part of your submission in class or in office hours; inability to explain your own work forfeits the corresponding marks (syllabus AI policy).

Late work. 10% penalty on submission after the deadline, a further 10% per additional 24 hours (syllabus).

The report — and the six tasks inside it

Your `.qmd` renders to a technical report with exactly these sections, in this order. Each section is one task below. The points on a task say how much of the 20 it carries; the rubric at the end says how the same 20 are split across the five things I grade (data, correctness, interpretation, code & report, reproducibility) — one set of marks, seen two ways. A section is R code, its printed output, *and* a paragraph saying what the output means — never output followed by the next heading (the rule Lecture 5 applies to itself). Every number you state in prose must appear, at the same rounding, in printed output somewhere in the report.

§	Heading	Lectures it draws on
0	Title, your name, one-paragraph abstract	—
1	The data: source, download, codebook	L1 (rows, columns, units), L3 (project, data frame)
2	Audit: extremes and shape	L1 (<code>sort</code> , <code>rank</code>), L3 (<code>apply</code>)
3	Correlation structure	L2 ($\bar{x}$, S , R , $ \mathbf{R} $), L4 (the noise floor, ordered <code>corrplot</code>)
4	Displays	L4 (<code>ggpairs</code> , colour, the pairwise fallacy)
5	Distances and unusual units	L5 (the workflow, <code>dist</code> , <code>scale</code> , <code>pheatmap</code> , <code>mahalanobis</code>)
6	Conclusions and limitations	all five, in words
A	Appendix: <code>sessionInfo()</code>	L3

Task 1 (Discover and build — report §1 and the CSV; 5 pts)

- Name the source, the exact tables or indicator codes, and the date of download (type it). Show the download code. State what one **row** is.
- Give a **codebook table**: for every column, its name in your matrix, what it measures, and its **unit**. Mark which columns share a unit. Say why *these* columns: what question about the rows would make a statistician put them side by side? (Two sentences; “they were available” scores zero.)
- Print `dim()`, `str()` and the per-column count of missing values *before* you drop anything. Say how many rows you lost to missing values, and to what else (aggregates, a total row, an age filter, a suppressed cell), each with its count. Print the final `dim()`. Confirm $n \geq 40$, $p \geq 8$, two units.
- Print your own unit of interest — Qatar, a state, a participant — as one row of the matrix, and keep it: Task 4 follows it.

Report: the download code, the codebook, both `dim()`s and the accounting between them.

Task 2 (Audit — report §2; 3 pts)

- One table, built with `apply()`: for each column its minimum, median, maximum and standard deviation. Name the row that holds each column’s maximum (`rownames(X)[which.max(X[, k])]`) for the two columns whose maximum is farthest from the median in units of the SD, i.e. the largest $(\max - \text{median})/\text{SD}$.
- Report **one thing about the columns that this table does not show**, with the output or figure that made you notice it (shape: `hist()`, or `sort()` of the column with the names kept). It will matter in Task 5.

Report: the table, the two named rows, the finding in one sentence.

Task 3 (Correlation structure — report §3; 4 pts)

- Print $\bar{\mathbf{x}}$, $\mathbf{S}$ and $\mathbf{R}$ (rounded; keep your column names short when you build the matrix, so the $p \times p$ tables fit the page). State their dimensions before running `dim()`.
- The strongest correlation: the pair, the value, and the word “absolute”. Verify it a second way from $\mathbf{S}$ alone, $r_{jk} = s_{jk} / \sqrt{s_{jj}s_{kk}}$, pulling the three numbers out of `cov(X)` by name.
- $|\mathbf{R}|$, and one sentence on what its value says about how much of p variables’ worth of information these p columns carry.
- The noise floor for your n and p** (Lecture 4 §5): with `set.seed(481)`, simulate 500 tables of independent `rnorm` columns of your own $n \times p$, take the median of the largest $|r|$, and report how many of your $\binom{p}{2}$ pairs clear it. Then draw the `corrplot` with the variables in **an order you choose and state** — by theme, as Lecture 4 §5 did with `R[fam, fam]` — and say in the caption what the order is and what it lets the eye see.

Report: the three matrices, the pair and its verification, $|\mathbf{R}|$, the floor and the count, the figure in your stated order with a caption.

Task 4 (Displays — report §4; 3 pts)

Deliberately under-specified: you choose the columns.

Draw **one** `ggpairs()` on a subset of **four to six** columns you choose (Lecture 4: the default display, unreadable beyond $p \approx 8$). If $n > 500$, draw it on a seeded random sample of 400 rows (`set.seed(481)` then `sample()`) and say so in the caption. If your matrix has a label, map it to `colour`. Then answer, citing panels by row and column:

- Name **one thing this figure shows that R cannot**: curvature, a cluster, a unit far from the cloud, or — if you have a label — a pooled correlation that differs from the within-group ones. Give the panel and the numbers.
- Say why you chose *these* columns and not the others, using your Task 3 numbers.
- The pairwise fallacy** (Lecture 4 §7): in one sentence, what kind of unusual unit can this figure not show you?

If $n \leq 40$ after a subset you can defend, you may add one row-side display (`stars` or the data heat map); it must carry a caption saying what the arrangement of variables is and that it is the analyst’s choice.

Task 5 (Distances and unusual units — report §5; 5 pts)

Deliberately under-specified: you choose the distance and the subset, and you defend both with the decision table of Lecture 5 §4.

- Step 1 of the workflow**: print `round(apply(X, 2, sd), 2)`. How many orders of magnitude separate the largest and smallest spread? Decide — and say in one sentence — whether the plain Euclidean distance is usable on these columns.
- Your unit’s neighbours**. For the unit you kept in Task 1(d), print its three nearest rows under (i) Euclidean on the raw columns, (ii) standardised Euclidean, (iii) the Mahalanobis distance from that unit (`mahalanobis(X, center = X[u, ], cov = cov(X))`, which returns d^2 — compare *orders*, not values). If n is in the thousands do not build `as.matrix(dist(X))`; Lecture 5 §4’s line with your unit in place of the centre, `sqrt(rowSums(sweep(X, 2, X[u, ])^2))`, gives every row’s distance from it. Do the three lists agree? Explain every disagreement by naming the column that causes it (Lecture 5 §3).

- (c) **One heat map.** Choose a subset of **about 40 rows or fewer** you can defend (a region, an income group, a seeded random sample, the 40 nearest to your unit ...; Topic C may keep all 52), choose *one* distance from the decision table, compute it with `dist()`, and draw it with `pheatmap()` exactly as Lecture 5 §2 does: `cluster_rows = FALSE`, `cluster_cols = FALSE` and **one fixed row order you state** (your unit first, then by distance from it, is the natural one). Say in the caption which distance, why, and what the order is. Name the closest pair and the row that is far from everybody.
- (d) **The unit no single column flags.** Compute the Mahalanobis distance of every row from the centre. Print the five largest d^2 and, after them, the five largest *standardised Euclidean* distances from the centre ($\sqrt{\sum_k z_{ik}^2}$, i.e. `sqrt(rowSums(scale(X)^2))`, Lecture 5 §4's line on the standardised table). For each row that is in the first list but not the second, print its standardised row and say which *combination* makes it unusual when no single $|z|$ does. If the lists agree entirely, say which single column is doing the work for the top row, and why d^2 cannot then add anything.
- (e) **The wall.** Print `det(cov(X))` and `dim(solve(cov(X)))`, and state n/p . Then add to a copy of your matrix one column that is the *sum* of two existing columns, run `mahalanobis()` on the copy, and report what happens to $|S|$ and to the call — and why (Lecture 5's ending).

Report: the SD line and the decision; the three neighbour lists with the disagreements explained; the heat map with its caption; the two top-5 lists and the standardised rows; the determinant, the n/p sentence, and what the added column did.

Task 6 (Conclusions and limitations — report §6, the abstract, and the appendix; no separate points)

No new numbers. Write, for a reader who knows the subject (a health economist, an epidemiologist, an energy analyst) and no multivariate statistics: (a) in 3–5 sentences, what the data matrix says about its rows — which columns move together, which unit is unusual and *why*, in the right units; (b) **two limitations** specific to *your* build (the year mix, the merge losses, age adjustment, one country's value setting a column's SD ...); (c) the sentence you would say if the reader asked “so if a state cut X, would Y fall?” — **a causal verb applied to these data (“leads to”, “drives”, “improves”)** costs the whole of the interpretation criterion. Then write the abstract last, and end with `sessionInfo()`. This task carries no points of its own: it is where the *interpretation* and *code & report* criteria of the rubric are read.

Before you submit — tick every line

- | | |
|---|--|
| <ul style="list-style-type: none"> ☐ I closed R, reopened the project, put my CSV beside the <code>.qmd</code>, and rendered from a fresh session — it worked. ☐ The download chunk is shown (<code>echo: true</code>) and rebuilds my CSV when I run it alone. ☐ The analysis chunks start from <code>read.csv()</code> of my CSV; nothing comes from my console history or the internet. ☐ No absolute paths (<code>/Users/...</code>, <code>C:\...</code>). ☐ Every number in my prose appears in printed output. | <ul style="list-style-type: none"> ☐ Every figure has labels and a caption saying what to look at; the heat map's caption names its distance and its row order (clustering off). ☐ Row names are the unit's name or code, so distances print with names. ☐ <code>set.seed(481)</code> sits immediately before every <code>replicate()</code> and every <code>sample()</code>. ☐ No test, no p-value, no component, no “outlier”. ☐ All three files carry the required name; I acknowledged any AI assistance in one line at the end. |
|---|--|

Marking rubric (20 pts)

Criterion	Pts	Full marks	Zero
Data discovery & audit	4	A matrix built by the student from the source, $n \geq 40$, $p \geq 8$, two units, none from the course panel; the accounting from raw rows to final n is complete; the codebook has units.	The CSV cannot be rebuilt; aggregates or a total row sit among the units unnoticed; a column from <code>whwide2022</code> .
Correctness	5	$\bar{x}$, S , R , $ \mathbf{R} $, the floor, the three neighbour lists, the two top-5 lists and the heat map all computed correctly on the submitted CSV; the S -route verification goes through <code>cov()</code> .	Numbers that no output supports; a “verification” that calls <code>cor()</code> twice; <code>dist()</code> on unscaled columns with a 300-to-1 spread ratio and no comment.
Interpretation	5	Every output is followed by what it means; the “ R cannot show” panel, the neighbour disagreements and the “combination, not a column” row are each explained with numbers; §6 names units, a real limitation, and refuses the causal reading.	Output followed by the next heading; generic sentences (“corrplot is a good way to show correlations”); any causal claim from these observational data; “outlier” used as if defined.
Code & report quality	3	Download and analysis chunks separated; readable, commented, one idea per chunk; only functions from the list on page 3; the report has the seven sections in order; figures captioned.	Unrunnable or uncommented; an untaught package or method used without a reason; sections missing or in a different order.
Reproducibility	3	The <code>.qmd</code> renders from a fresh session beside the CSV and produces the submitted PDF; seeds set; <code>sessionInfo()</code> at the end.	Does not render, or depends on objects, paths or downloads that exist only on the student’s machine.
Total	20		

Grade returned as: data ____/4 + correctness ____/5 + interpretation ____/5 + code & report ____/3 + reproducibility ____/3 = ____/20.